

Strain values

- Maximum Flexibility
- Percentage Elongation
- Percent offset

Energy values

- Resilience
- Toughness
- Fracture Toughness
- Impact strength

Failure of the material below yield strength

- Impact strength
- Fatigue strength
- Creep → Permanent deformation

Fracture

Resilient Material

- High Proportional limit
- Low modulus of elasticity

Tough Material

- High Proportional limit
- High Percentage Elongation

Brittle Material → Tough عمرها متبقي

Proportional limit → Material or strong weak

Modulus of elasticity → Material or Flexible Stiff (Rigid)

Percentage Elongation → Material or Ductile (Malleable) Brittle